A COMPREHENSIVE LITERATURE REVIEW IN ADVANCEMENTS AND CHALLENGES IN DIAGNOSIS OF LUNG CANCER

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Abstract:

Despite significant advancements in healthcare, the lung cancer diagnosis remains a challenge, for enhancing lung cancer detection and the patient outcomes, AI and DL has techniques that has emerged as promising tools. One of the deep learning methods, Convolutional neural network used in lung cancer detection and shows the good accuracy and the ensembling method used to compare the predictions of two or more models and explainable AI is used for the better understanding of the clinicians, even the clinical implementation, the robustness of the model, bias remains challenging. This literature review explores the recent advancements, challenges and the future directions in AI and DL applications in lung cancer diagnosis.

Introduction:

In recent days, a lot of improvements are happening in the Healthcare field, even though lung cancer is still one of the deadliest forms of cancer in the world and it is very much necessary for us to make continuous advancements in diagnosis of lung cancer for the early detection and for the treatment too. The usage of Artificial Intelligence and Deep Learning techniques is doing good in the Advancement of lung cancer diagnosis.

Amidst the multitude of cancer types affecting individuals daily, lung cancer stands out as one of the most prevalent and frequently diagnosed malignancies globally. For instance, in the United States alone, (Lung Cancer Research Foundation (LCRF),2023), an estimated 2.3 million cancer cases were reported in 2023, affecting approximately 1 in 16 men and 1 in 17 women.

The work of Shah et al., (2023) implemented the two-dimensional Convolutional Neural Network and it gives a good result by using the ensembling methods. The variations in the datasets like the Ethnicity, genes need different architecture of the model and what is the trust level of the clinicians and the patients, were explained by De Margerie-Mellon, C. and Chassagnon, G. (2023) in their article.

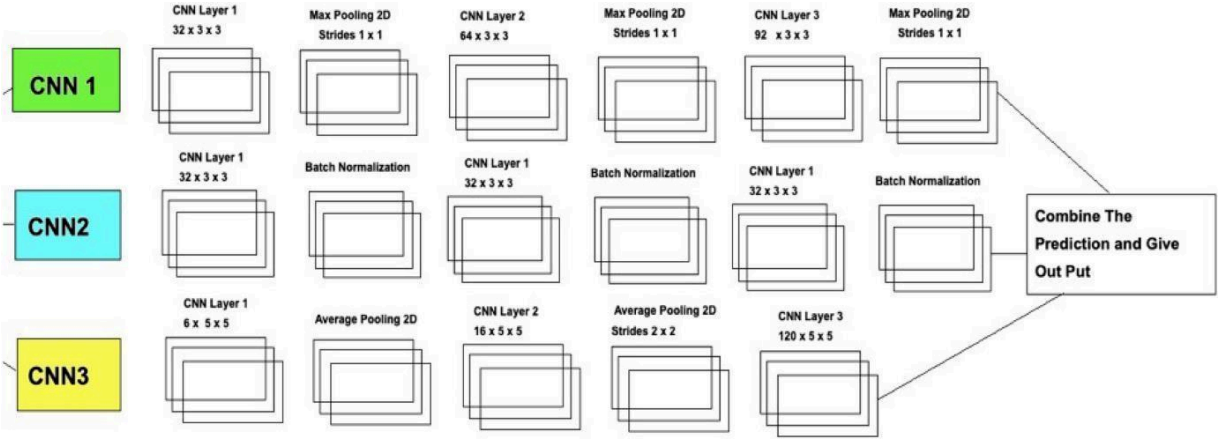

Figure: 1, Shah et al., (2023), *Deep learning 2D approach towards the detection of lung cancer.*

The robustness of the model will lead to the misperformance of the model, the usage of adversarial Images for the training of the model to overcome the robustness and to give good accuracy was explained by Joel M.Z et al., (2022) in their article. For using AI and DL in clinics for the detection of lung cancer, clinicians need to know the inner workings of the AI and DL model, the article from Waymel Q et al., (2019) has a brief statement about this.

Challenges in Clinical Implementation:

The greatest challenge of implementation of Artificial Intelligence (AI) tools for the diagnosis of lung cancer as per De Margerie-Mellon, C. and Chassagnon, G. (2023) is the heterogeneity of the published studies. Not all the datasets have the same data, there will be some differences like ethnicity, genes, etc., and every different dataset wants different types of algorithm architecture, these variations make it challenging to combine the information from multiple sources and another important challenge of the clinical implementation is a comparison of the radiologist with the machine, how well the machine functions over the time.

The next level of challenge is the obscure inner workings of the predictive deep learning models, if clinics are using Deep learning applications or techniques for the detection of lung cancer nodules, the clinicians must be confident about how that AI and DL model works and again finding the threshold performance for the clinical use is so tough. As De Margerie-Mellon, C. and Chassagnon, G. (2023), before implementing the AI and DL applications in clinics, the clinicians must know something about the model's inner workings.

The next thing is the robustness of the model, as per Joel M.Z et al., (2022) robustness is something that is the ability of a model to perform consistently and accurately even when faced with various challenges and uncertainties. Robustness in DL models involves their capacity to handle variations in data, noise, and unexpected inputs without significantly compromising performance. The adversarial training technique is used to prevent the negative effects of the adversarial images.

When using the ensembling methods, if every other model is trained from different subsets of data, the accuracy of the model will lead to bias, according to Albahri (2023), bias is something that is systematic deviation or error because the trained subset of data is different in every other individual constituent model.

The results from the radiologist and the AI models is not showing much difference. Comparatively, the impact of the AI model over the radiologist for decision-making is largely unknown, and AI has the potential to be treated the same as radiologist for its workflow and again a much more careful evaluation is required for the clinical practice.

Critical Analysis:

The brief above on the challenges in the diagnosis of lung cancer notably focuses on the combination of artificial intelligence and deep learning innovations. It gives a comprehensive overview of the current happenings in the world. The review Emphasizes the several solutions of AI in enhancing diagnostic accuracy. It also shows the critical challenges.

First, the review acknowledges the transformative potential of AI and DL in medical imaging, specifically in the detection of lung cancer nodules. The discussion from the article of Shah et al., (2023) on the application of the two-dimensional approaches and ensembling methods shows the outstanding paces of achieving a unique level of accuracy in lung cancer diagnosis. By spotlighting specific examples and technologies, the review productively illustrates the impact of AI on current diagnostic practices.

The ensemble method is used to know how fairly the accuracy is given by the separate models, whereas the work of (Bokefode, Rao, and Kumaraswamy (2022)) shows it works on the concept of voting technique, where voting is defined as comparing the predictions of all the models and the challenge here

is the bias of the model and it rises from the trained data of separate models. From

the work of Hofmanninger et al., (2020), to overcome the biases of the model in the ensembling method, the training of each model should be done by dataset having a diverse of data. Here, they are not mentioning that the dataset for all the models should be the same, but the dataset for every other model should need to have a diverse subset of data.

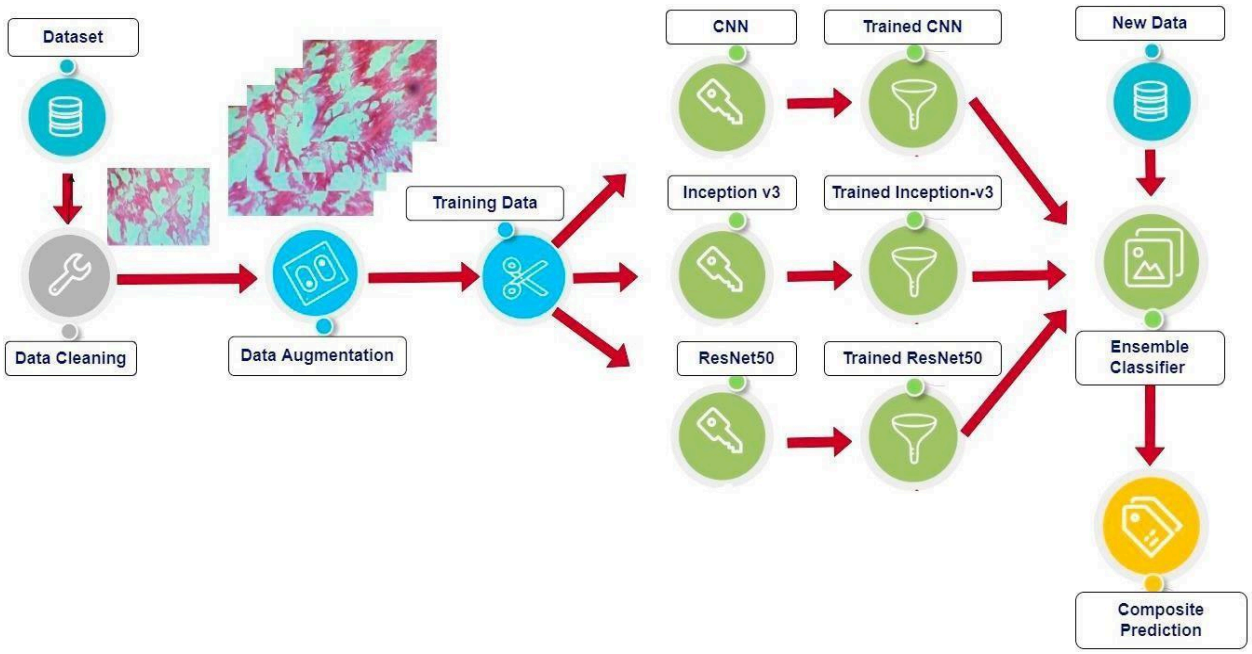

Figure: 2, Bakefode, Rao, Kumaraswamy (2022), *Ensemble deep learning methods for lung cancer diagnosis*, Elsevier.

The most important challenge as of De Margerie-Mellon, C. and Chassagnon, G. (2023) is the heterogeneity of published studies, which presents difficulties in synthesizing results and evaluating the generalizability of AI algorithms. Pointing up the need for greater standardization and clearness in AI research. The threshold performance of the model can be overcome by the hyperparameter tuning like changing the learning rate accordingly for the best performance (Zhang M et al., (2021)).

The robustness of the model will lead the model to give incorrect predictions (Chen and Das, 2023). To overcome the robustness of the DL model, the technique used by Joel M.Z et al., (2022), is the usage of adversarial images. By training the model of both adversarial images and the normal images, the DL model learns to classify things and it shows higher accuracy compared with accuracy with the trained images of the normal sample.

Also, the review addresses the opacity of the predictive model, as per Waymel Q et al., (2019), before implementing the AI models in clinics, the radiologist must know something about the inner workings of the model, and clinicians trust and confidence to AI model wants to be good. The explainable AI(XAI) can be used for the understanding purpose of the clinicians or any other healthcare professionals, like how (Wani, Kumar, and Bedi (2023)) used DeepXplainer to explain the CNN for their own DL model. After the classification layer, the explainability layer is placed, which determines whether the patient is diagnosed with cancer or not. This layer gives reason for outputs. Finally, these explanations are passed to the medical practitioner layer, where the clinicians understand these explanations for the outputs and the clinicians will be able to explain to the patients why they have been diagnosed with cancer or why not.

Ethical Considerations:

Ethical considerations in the AI application used for the detection of lung cancer, as per the statement of De Margerie-Mellon, C. and Chassagnon, G. (2023), if the dataset used for training the AI model does not have the data of a diverse population, this could lead the AI application to give the wrong detective results.

Day by day the role of Artificial Intelligence in the radiology field is rapidly increasing in the diagnosis of lung cancer, leading to the displacement of jobs from healthcare professionals, around 20% of doctors all over the world are concerned about what if Artificial Intelligence took all the places of the radiologist and other healthcare professionals (DocWire News, 2023), are we able to expect the expertise results? No. This is one of the main ethical considerations.

The next thing is the reliability of the AI algorithms is very important for patient safety, if the AI application is doing the work of detecting cancer, we cannot expect how well the AI model is validated, if it's not validated properly, this leads to false positive and the false negatives in the diagnosis, according to Martin et al., (2023) this could lead a serious consequence for the patients.

In some instance, the lung nodules detected by deep learning algorithms is slow growing lesions that will be never able to become cancer lesions according to the clinicians, but the current tests lack the ability to find the difference between indolent and aggressive lesions, all the detected nodules

are treated as potential cancer nodules, this will lead to overdiagnosis (Han S.S et al.,2017).

The risk of the data being accessed by unauthorized organizations leads to violation of privacy regulations. The safeguard measures must be put in place to keep the privacy of the patients and we must make sure that the personal information of the patients is not disclosed without their consent (Khalid et al., 2023).

Future Direction and Recommendation:

Despite the challenges, the future of AI tools in lung cancer diagnosis appears promising. The main important thing to enhance the comparability across research efforts is we must set a mandatory guideline for building a standard Deep Learning model and very crucial to have an availability of larger and high-quality datasets while training the AI model the overcome the robustness.

When every clinic implements an AI model for the detection of lung cancer nodules, Waymel Q et al., (2019) say that radiologist plays an important role in the deployment of the AI tools, and they should always want to keep up-to-date information about the alignment of the AI tool with clinical needs and workflows.

At the same time, the advent of AI-based pathological analysis is offering a lung cancer diagnosis technique that almost gives a precise result. The imaging techniques like computed tomography (CT) compared with AI facilitate early detection and find the difference between benign and malignant tumours. The Future Perspective article of Huang, S. et al. (2023). Even after all the careful consideration, the fact of trusting the AI models with the radiologist is comparatively low.

Conclusion:

In conclusion, AI and DL hold immense potential for transforming the diagnosis of lung cancer, the significant advancement, several challenges, including heterogeneity of studies, algorithm transparency, the threshold of the model performance, and integration of it in the clinics need to be addressed for the successful implementation. In terms of ethical considerations, the first thing to be addressed is whether it can be used for the diverse population of the world, and the second thing is we must make sure that AI is not displacing all the jobs of healthcare professionals like radiologists and other clinicians.

Future research efforts should focus on standardization, validation, interdisciplinary collaboration, and standard guidelines for creating and training an ethical AI and Dl model to get the full benefits of AI in improving lung cancer outcomes and the mortality rate.

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